Detecting Misogyny in Spanish Tweets. An approach based on Linguistics Features and Word embeddings

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Abstract

Online social networks allow powerless people to gain enormous amounts of control over particular people's lives and profit from the anonymity or social distance that the Internet provides in order to harass other people. One of the most frequently targeted groups comprise women, as misogyny is, unfortunately, a reality in our society. However, although great efforts have recently been made to identify misogyny, it is still difficult to distinguish as it can sometimes be very subtle and deep, signifying that the use of statistical approaches is not sufficient. Moreover, as Spanish is spoken worldwide, context and cultural differences can complicate this identification. Our contribution to the detection of misogyny in Spanish is two-fold. On the one hand, we apply Sentiment Analysis and Social Computing technologies for detecting misogynous messages in Twitter. On the other, we have compiled the Spanish MisoCorpus-2020, a balanced corpus regarding misogyny in Spanish, and classified it into three subsets concerning (1) violence towards relevant women, (2) messages harassing women in Spanish from Spain and Spanish from Latin America, and (3) general traits related to misogyny. Our proposal combines a classification based on average word embeddings and linguistic features in order to understand which linguistic phenomena principally contribute to the identification of misogyny. We have evaluated our proposal with three machine-learning classifiers, achieving the best accuracy of 85.175%. Finally the proposed approach is also validated with existing corpora for misogyny and aggressiveness detection such as AMI and HatEval obtaining good results

Keywords: Misogyny Detection, Text Classification, Natural Language Processing, Machine-Learning

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1. Introduction

A survey carried out by the Pew Research Center[1] showed that roughly four-in-ten Americans have experienced online harassment [1] in social media. Certain groups are more likely to experience some sort of trait-based harassment than others, the most important being those motivated by ethnicity or gender. This survey revealed that women were about twice as likely as men to state that they had been targeted as a result of their gender. Women, and specifically young women, usually undergo sexualised forms of harassment. For example, 21% of women between 18 and 29 years of age report being sexually abused online, more the double that experienced by men; and more than half of them claim to have received explicit images that they had not requested.

According to Kate Manne in "The Logic of Misogyny" [2], misogyny refers to social environments in which "*women will tend to face hostility of various kinds because they are women in a man's world who are held to be failing to live up to men's standards*". Manne refers to a relevant difference between the concepts sexism and misogyny, as sexism refers exclusively to a discriminatory behaviour between men and women, whereas misogyny creates the artificial terms of *good* and *bad* women with the aim of punishing what misogynists consider to be bad women. In *Woman-Hating: On Misogyny, Sexism, and Hate Speech*, Louise Richardson-self identifies misogynous behaviour with signs of hostility with a coercively function [3]. In this respect, it is possible to distinguish among hate speech discourses that oppress by including certain signs of harassment or intimidation towards stigmatised group members. For example, much of the damage done by the spread of gossip and slander on the Internet causes harm to specific groups. Women specifically suffer from a significant portion of this harmful material that feminists call *objectification*, which refers to being considered objects for men's use and abuse. Moreover, it is possible to find misogyny hidden in messages with the aim of discrediting someone because of their gender, attempting to lessen women's voices as regard an important issue, expecting the accomplishment of certain unfounded stereotypes towards women or even giving women needless explanations, usually in a condescending manner *(aka mansplaining)*, which typically also involves the chronic interruption of women when they are speaking.

We have, in recent years, witnessed the overgrowing usage of social media, in which freedom of expression is subject to basic rules of behaviour in order to guarantee a healthy environment and prevent online harassment. However, the Internet has allowed powerless people to gain enormous power over the lives of particular people by using specific properties on the Internet such as anonymity [4]. Women who have attained success consequently suffer from sexists attacks, thus originating the term known as Violence Against Women in Politics (VAWIP) [5, 6]. Online misogyny has, therefore, been compared to witch-hunting, since both have a similar function: coercing women in order to

[1] `https://www.pewresearch.org/`

prevent them for progressing [7].

Social Computing and Sentiment Analysis technologies have a great potential in order to extract critical insights on opinions shared over social networks which can help to identify hate speech and discrimination [8]. These technologies have been applied in multiple text classification tasks such as irony [9] or hate speech detection [10]. If we consider misogyny to be a form of hate speech, then hate speech detectors should perform perfectly well when analyzing text containing misogynous traits; but in some cases, misogyny is very subtle and deep and can, therefore, be difficult to distinguish. Moreover, context and cultural differences can complicate this identification to an even greater extent. Despite these difficulties, automatic misogyny identification is gaining relevance as a challenging task, and has consequently been proposed in conferences and workshops related to Natural Language Processing (NLP) owing to its social relevance. However, to the best of our knowledge, there are only two datasets available in Spanish for misogyny identification, signifying that more resources in Spanish are still required for a better understanding misogynous behaviour.

The main contributions of this work can be summarized as:

- **Spanish MisoCorpus-2020**. The compilation and classification of a balanced corpus of tweets related to misogyny written in Spanish. This corpus is subdivided into the three main subsets: (1) Violence Against Relevant Women (VARW), which concerns misogynistic messages directed towards specific women who have gained social relevance, (2) European Spanish vs that of Latin America (SELA), which concerns misogynistic messages written in Spanish from Europe and Spanish from Latin America, with the aim of understanding how misogynous content is affected by cultural background, and (3) Discredit, Dominance, Sexual harassment and Stereotype (DDSS), which concerns tweets containing general aspects related to misogyny, such as derailing, rape, or gender violence, among others. The combination of these subsets is the Spanish MisoCorpus-2020, which has been released for use by the scientific community.

- **The development and evaluation of a machine-learning model with which to detect misogyny**. This model is capable of identifying tweets written in Spanish based on linguistic features and word embeddings. During the evaluation of the model, we analysed each subset of features with the whole corpus and each subset. The results were compared by applying three machine-learning classifiers: (1) Random Forest (RF), a decision tree classifier; (2) Sequential Minimal Optimization (SMO), a Support Vector Machines classifier (SVM), and (3) Linear Support Vector Machines (LSVM).

The remainder of the paper is organised as follows: Section 2 describes recent works and approaches concerning the detection of misogyny. Section 3 describes the way in which the corpus was compiled and its composition. Section 4 describes the linguistic features employed in our model along with the results

obtained from the experiments performed. Section 5 contains a detailed analysis of the results and, finally, Section 6 summarises the conclusions of the paper and presents future research directions.

2. Related work

In recent years, emerging technologies in Social Computing such as Sentiment Analysis (SA) have provided the capacity to understand citizens' attitudes and feelings which has promoted some domains such as marketing or customer support. For example, in [11] the authors combined Google Trends and Twitter enabling decision makers to monitor social networks in real-time as the literature has demonstrated the relationship among Google Trends to real-world phenomena. This new capacity of analysis in real-time can help to build safer places for the society by tracking on the Internet terrorism-related messages as well as other forms of harassment such as bullying or sexual harassment [12].

Automatic misogyny detection has, owing to its relevance, been proposed as a task in workshops such as IberEval[2], organised by the Spanish Society of Natural Language Processing (SEPLN), or EvalIta[3] organised by the *Associazione Italiana di Linguistica Computazionale*. Both workshops covered the identification of misogynous messages on Twitter in English, Italian, and Spanish. These tasks are analysed in detail owing to their relevance as regards our proposal.

IberEval organised the Automatic Misogyny Identification (AMI) [13] task with two available corpora, one in Spanish and one in English. The participants in the task could choose either of both of them. There were two sub-tasks for each corpus: (1) a binary classification for misogyny detection, and (2) a multi-class classification of different misogynistic traits, namely, stereotypes (aka objectification), dominance, derailing, sexual harassment, and discredit. In addition, tweets from both corpora were annotated as *active* or *passive*, according to whether or not each tweet was directed towards to someone in particular. The participants in this task employed different techniques in order to develop their models. They applied statistical and other features, such as n-grams, word embeddings, and linguistic features. With regard the machine-learning classifiers, the participants principally employed Support Vector Machines whereas others applied deep-learning based techniques. An overview of this task can be consulted at [13], which contains a more detailed description of the participants and the results.

As occurred in IberEval, EvalIta also organised an AMI task [14], but in this case the Spanish dataset was replaced with one in Italian and a new corpus (as regards both training and testing) was compiled for the English dataset. The other characteristics of the task were similar, signifying that the participants were asked to solve a binary classification problem, and a multi-class

[2]https://sites.google.com/view/ibereval-2018/home
[3]http://www.evalita.it/2018

classification problem. The ten participating teams in EvalIta employed similar techniques, such as Bag of Words, char-grams, or lexical features, in order to improve the accuracy of their models and provide a better understanding of the results. An overview of this task can be consulted at [14].

The relevant literature contains not only the aforementioned tasks, but also more research regarding misogyny identification, although principally for the English Language. For example, Frenda et al. [15] carried out an in-depth analysis of misogynistic and sexist comments in order to identify differences and analogies between misogyny and sexist behaviour towards women. In their analysis, they compared the English datasets employed in the AMI IberEval [13] and the AMI EvalIta [14], with the NAACL_SRW_2016 corpus regarding Hate Speech in a Twitter annotation [16] corpus, in which tweets were annotated as *sexist*, *racist*, or *none*. Their analysis indicated that sexism and misogyny had similar linguistic features, such as the words related to Sexuality, Femininity, or the Human Body, whereas the number of offensive words was more frequent in the misogynistic corpora.

Another work regarding misogyny identification can be found at [17], where Lynn et al. made use of the Urban Dictionary[4] to compile slang words related to misogyny in order to compare the performance of two deep learning techniques: Bidirectional Long Short Term Memory (BiLSTM) and Bidirectional Gated Recurrent Units (BiGRU), which outperformed conventional machine learning techniques. On the other hand, Tanvi Banerjee et al. [18] developed a machine learning classifier capable of identifying topics concerning Gender-Based Violence (GBV), by analysing tweets from a linguistic perspective in order to detect whether the writer was expressing beliefs, queries, metaphors, or was reporting facts. They analysed supervised and unsupervised machine learning models in order to determine the influence of the features on the belief class.

3. Corpus

This study has been carried out using a dataset concerning misogynous behaviours. We first conducted a review of the bibliography to find existing corpora related to misogyny in Spanish. There are, however, few available public resources regarding automatic misogyny identification, as is reported in the survey carried out by Shushkevich and Cardiff [19]. This survey describes, among other things, the Spanish AMI 2018 dataset from IberEval, described at [13] (see Section 2). This corpus was compiled from (1) tweets with representative keywords related to misogyny, (2) tweets from potential victims' accounts, and (3) tweets from users identified as misogynist. This corpus was compiled in 2017 over a period of 4 months and was manually annotated by two annotators and a judge. The training set is almost balanced and contains 3,307 tweets. There is also another related corpus available at HatEval Spanish corpus,

[4] https://www.urbandictionary.com/

which is from the SemEval 2019 Task 5 and concerns the identification of hate speech directed towards immigrants and women [20]. This corpus is composed of 6,600 tweets and it is not balanced, with around 58% of tweets labelled as *Hateful* as opposed to those tweets labelled as *Non-Hateful*. Although this corpus is a good resource, only 1,664 tweets were classified as misogynous, while the rest were non-misogynous or were related to hate speech against immigrants.

However, these corpora have limitations. First, neither corpora distinguishes between tweets compiled from different Spanish-spoken countries and it is not, therefore, possible to analyse misogyny according to the cultural background from which it originates in order to compare oppressive messages from different variants of the Spanish language. Second, although AMI distinguishes between tweets directed towards one person or collective, it is not possible to analyse misogynous messages directed towards relevant women. We, therefore, decided to compile a new corpus in order to overcome these limitations and to provide more linguistic resources for misogyny detection in Spanish.

The new corpus presented in this paper is formed of tweets labelled as *misogyny* or *not misogyny*, and it is divided into the following subsets:

- **Violence Against Relevant Women (VARW)**. This corpus contains tweets directed towards relevant specific women. For example, we compiled tweets addressed to the young environmental activist Greta Thunberg, who participated in the COP25 (2019 United Nations Climate Change Conference) that was celebrated in Madrid (Spain). The Spanish newspaper "El País"[5] analysed 440,000 tweets (period: November 20th to December 11th, 2019) that mentioned Greta Thunberg in their comments. The authors of this study used the tool *Genderizer* [21] to determine which users were men or women. They discovered that men performed around 2/3 of all the tweets and that nearly 14,000 of the tweets contained misogynist adjectives such as *histérica* (hysterical), *estúpida* (stupid), or *gilipollas* (asshole) among other insults.

  In addition, other accounts in which tweets were compiled where those of female politicians. As Spain was repeating her elections, as a result of previously failures in the formation of a government, there was a great deal of activity in social networks directed towards politicians. We consequently selected the accounts of women politicians from the left-wing and right-wing Spanish parties, in addition of those of journalists and other relevant women. We also attained tweets from news site accounts related to these women in order to have objective tweets which reported cases but that could not be considered as misogynous.

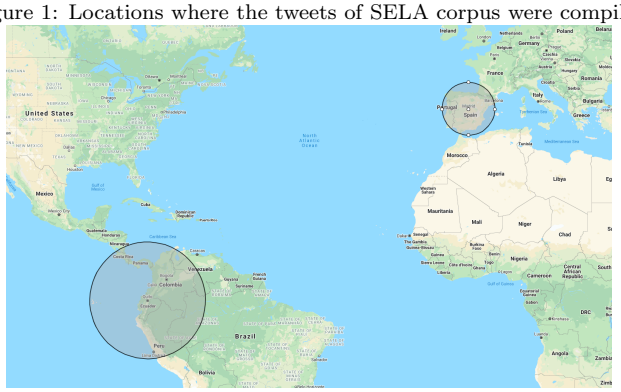
- **European Spanish vs that of Latin America (SELA)**. Upon analysing the results of the AMI Task [13], our attention was drawn

[5] `https://elpais.com/tecnologia/2019/12/12/actualidad/1576106658_165755.html` (accessed: at April 28, 2020)

to the fact that the overall accuracy of the runs was higher in English that in Spanish, with the best accuracy of 91.3% for the English dataset and a best accuracy of 81.4% for the Spanish dataset. An analysis of each corpus revealed that tweets compiled in Spanish were written by European Spanish people and Latin American people. Although all languages have differences based on their cultural background, Armstrong et al. [22] noticed that European and Argentinian Spanish differ considerably in terms of the relative meaning frequencies of their constituent homonyms. We, therefore, decided to compile tweets from Latin American and European Spanish people separately. The tweets from Latin America were obtained from `latitude: -0.1596997, longitude: -78.452125313, radius: 1,500 km` whereas the Spanish tweets were collected from `latitude: (40.416705, longitude: -3.703583, radius: 520km`. Figure 1 shows the areas from which where the tweets were compiled.

- **Discredit, Dominance, Sexual harassment, and Stereotype (DDSS)**. We extracted tweets from keywords related to misogynistic traits, such as discredit, dominance, sexual harassment or stereotype. We mainly selected those tweets that discussed topics related to gender violence. In addition, and as a consequence of the Spanish electoral debate, there was a discussion in which certain political parties dismissed the concept of *gender violence* in favor of the term *violencia intrafamiliar* o *violencia doméstica* (domestic violence) with the aims of dismissing the visibility of *gender violence* [23].

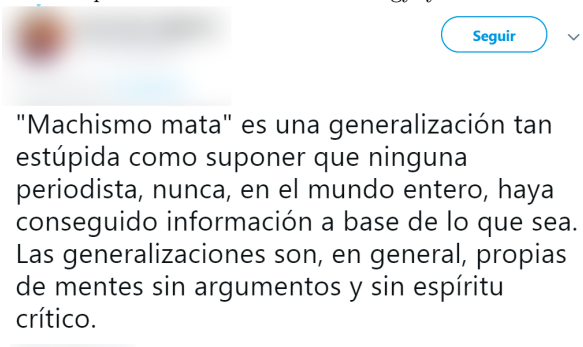
Figure 1: Locations where the tweets of SELA corpus were compiled



We followed the same approaches employed in IberEval and EvalIta and used Twitter as a data provider. We compiled around 32,969 tweets from the criteria previously described for each subset of the corpus. Figure 2 and Figure 3 show two examples from the corpus, which reflect the complexity of misogyny identification. The first example (Figure 2) concerns about the expression *"machismo mata"* (male chauvinism kills), which alludes to the fact

that intolerant behaviour towards female individuals result directly or indirectly in the death of the victims. However, the author of this tweet reduces the intensity of this expression by labelling it as a stupid generalization. Moreover, he suggests that women journalists take advantage of their physique for the benefit of their profession. In the second example (Figure 3), the author gives some insights into the fact that people relativize the age of the victims of harassment according to their own interests. In particular, the author complains about people who believes that Greta Thunberg should not be taken seriously because of her age; however, when the same people discuss a recent case of the sexual abuse of a child a few months younger[6], they insist that she is totally capable of making adult decisions and this it should not be considered as assault (even when the age of consent is 16, according to Spanish legislation). Moreover, some newspapers change the way in which they refer to female victims according to their age, which leads to a depersonalization of the victims and causes a lack of empathy from the reader's point of view in order to polarize the audience [24].

Figure 2: First example of a tweet labelled as misogyny in the MisoCorpus-2020[7]



We then followed a manual supervision approach to classify the tweets in a binary classification: *misogyny* or *not misogyny*. However, all the tweets captured from news sites were automatically labelled using a distant supervision approach [25] as *not-misogyny* because we agreed that all had an objective point of view. The manual classification of the tweets was performed by seven people, five of whom were women and two of whom were men. They were the authors of the manuscript, two colleagues from the University of Murcia, in addition to a Master's degree student. We considered the following guidelines when labelling the corpus:

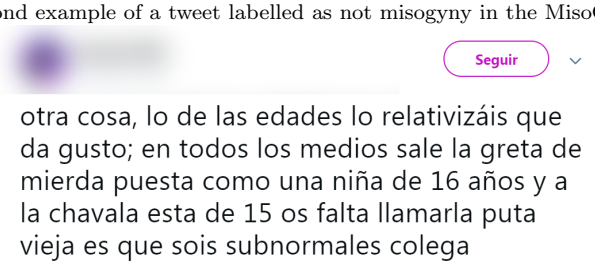
[6] `https://elpais.com/politica/2018/01/08/actualidad/1515439892_439672.html` Accessed: April 24, 2024

[7] In English: "Machismo kills" is a generalization that is as stupid as supposing that no female journalists in the world have ever obtained information by using the fact that they are women to do so. Generalisations are generally typical of minds without arguments or critical spirit.

- If the tweet contained images or links, and they were necessary to understand the meaning of the tweet, the tweet was labelled as *out-of-domain*.
- If the tweet contained words in the Catalan or Basque languages[8], the tweet was labelled as *out-of-domain*[9].
- If the tweet contained figurative language that made the motivation of the text unclear, the tweet was labelled as *out-of-domain*.
- If the tweet contained certain traits of misogynous behaviour, such as objectification, dominance, derailing, sexual harassment, or discredit, the tweet was labelled as *misogyny*.
- If the tweet contained a clear predisposition towards equality between men and women, even if slag or offensive language were employed, the tweet was labelled as *not-misogyny*.
- If the Twitter account belonged to an online news site, the tweet was labelled as *not-misogyny*.

The corpus was then processed to be transformed into a binary classification task (misogyny, not misogyny). We first removed those tweets labelled as *out-of-domain*, after which we selected the representative misogynous tweets as being those that had been annotated as *misogyny* by at least two or more annotators and which had not been labelled as *non-misogyny* by any of the annotators. That is to say, a tweet labelled as *misogyny* by one annotator and labelled as *non-misogyny* by another was discarded. Furthermore, the following guidelines were applied in order to select representative non-misogynous tweets: (1) Tweets compiled from news sites were automatically annotated as *not misogyny*, and (2) tweets labelled as *not misogyny* and *out-of-domain* were considered to be *not misogyny* tweets.

Figure 3: Second example of a tweet labelled as not misogyny in the MisoCorpus-2020 [10]



[8]Languages spoken in Catalonia and Basque country, autonomous communities in Spain

[9]This decision was made on basis that no linguistic models were available for these languages.

The Spanish MisoCorpus-2020 is composed of 7682 tweets related to misogyny written in Spanish. The statistics concerning the corpus are shown in Table 1. Note that the VARW, SELA and DDSS corpora are not disjoint datasets, signifying that the sum of the tweets contained in each subset is not equal to the total size of the MisoCorpus-2020 because some tweets may appear in different subsets. Each tweet was rated individually by at least two annotators. (1) The VARW dataset has been rated an average of 2.1529 times; the SELA dataset an average of 2.3076, the DDSS dataset 2.1595 and, finally, the MisoCorpus-2020 2.2240. We calculated the inter-coder reliability by applying the Krippendorff's Alpha, which measures the agreement from 0 to 1, where 0 is perfect disagreement and 1 is perfect agreement [26], and obtained an alpha of 0.6864. This value is quite low owing to the ambiguity of the domain, although Krippendorff suggested that tentative conclusions with an alpha higher than .667 are still acceptable.

Table 1: Corpus statistics and classification

Name	Misogyny	Not misogyny	Mean of annotations
VARW	2094	2094	2.1529
SELA	2081	2081	2.3076
DDSS	1665	1665	2.1595
MisoCorpus-2020	3841	3841	2.2240

We shared the compiled corpus with the community; however, according to Twitter guidelines[11], we could provide only the list of the IDs of the tweets, thus enabling the authors of the tweets to maintain the rights about their content on the Internet. The list of Twitter IDs of the whole corpus, as well as the Twitter IDs of VARW, SELA, and DDSS subsets are available at the following URL: `https://pln.inf.um.es/corpora/misogyny/misocorpus-spanish-2020.rar`. Each subset is divided into two files, one containing the IDs of the tweets annotated as misogyny and other file with the tweets labelled as not misogyny.

4. Evaluation

In this section, we detail which features were extracted in order to train the models (see Section 4.1), and the evaluation of each model with the whole corpus and each subset (see Section 4.2).

[10] In English: Another argument. You relativize the ages of women at your whim. In all the media, Greta (*referring to Greta Thunberg*) is referred to as a 16-year-old girl, while in the case of the 15-year-old girl (referring to a specific case of rape that took place in Spain and whose victim was a minor), the only thing you don't do is call her an old whore. You are retarded

[11] https://developer.twitter.com/en/developer-terms/more-on-restricted-use-cases

4.1. Feature extraction

Prior to feature extraction was a normalization stage. This stage eased the extraction of both the statistical and the linguistic features. The pipeline consisted of: (1) transforming each tweet into lowercase letters, (2) removing blank lines and HTML tags, (3) removing mentions, (4) fixing misspellings and (5) removing continuously repeated symbols. However, we maintained the original statement and used it to extract certain linguistic features, such as the percentage of misspelled words. With regard to URLs, Twitter uses a short mechanism that generates reduced versions of URLs that contains the token `http://t.co` followed by an identifier. However, as the same URL can generate two different identifiers, this hinders the search for duplicated messages and we, therefore, decided to strip the identifier and represent URLS with the `http://t.co` token. Finally, in the case of the hashtag symbol, we remove it in the normalised version of the tweet but maintaining the word, signifying that, the hashtag *#feminista* becomes *feminista* (feminist).

We then used the pre-processed version of the tweet to generate its sentence embeddings. Sentence embeddings are composed by word embeddings, which are dense representations of words within a high dimensional space, creating clustering of words that are semantically similar [27]. Sentence embeddings can be represented as the Average of Word Embeddings (AWE) in the text. Sentence embeddings behave similarly to word embeddings, as they have the same main properties [28]. In this work, we applied the Spanish FastText model [29, 30], which contains pre-trained word embeddings from Common Crawl[12] and Wikipedia [13].

The Linguistic Features (LF) were extracted by employing a tool designed by our research team and inspired by Linguistic Inquiry and Word Count (LIWC) [31], a text analysis tool that is capable of extracting "*emotional, cognitive, and structural components present in individuals' verbal and written speech samples*" by identifying words that belong to specific psycho-linguistic topics. Our approach, UMUTextStats, contains 253 different linguistic features that are grouped into the following categories:

- **Figures of speech (FSE)**. This contains words and expressions related to figurative language. This set of features contributes towards identifying texts whose meaning differs from their literal meaning [32]. It contains regular expressions from online sources regarding *hyperboles*, *idiomatics-expressions*, *rhetorical-questions*, *verbal-irony*, *understatements*, *metaphors*, and *similes*. For example, the expression *besar el santo* (literally, "to kiss a saint") is a Spanish idiomatic expression meaning "to have a stroke of luck", but is not related to any feeling of affection.

[12] `https://commoncrawl.org/`
[13] `https://es.wikipedia.org/`

- **Pragmatics (PRA)**. These features capture how the meaning is transmitted. For example, we identify (1) expressive lengthening, a linguistic device that consists of repeating some letters of a word for emphasis [33]; (2) typographical features, in order to measure the number of words written in uppercase letters, which are commonly used to express shouting in social media, and (3) onomatopoeia, which consists of the phonetic imitation of the sound that it describes, such as animal noises.

- **Morphological features (MOR)**. These features capture components of the words, such as stems, prefixes, or suffixes. These features were selected because Spanish, as a highly influenced language, makes use of inflectional morphemes to denote multiple syntax and semantic meanings. We are of the opinion that capturing these features is useful as regards identifying misogyny, since Spanish contains suffixes that can, for example, make the stem word derogatory. Moreover, Spanish requires gender and number matching for grammatical concordance which can help to identify whether the passive subject of a statement is male or female.

- **Grammar and spelling mistakes (ERR)**. The objective of this set of features is to capture inappropriate style patterns, such as errors with accentuation rules, or the percentage of misspelled words. With regard to misogyny identification, this set of features can help to identify those people who do not pay sufficient attention to what they are writing, perhaps because they are distracted or in a bad mood.

- **Part of Speech (PoS)**. This set of features helps to organise words as regards their syntactic behaviour. There are features that capture the percentage of adjectives, pronouns or verbs within the texts. Moreover, it makes a considerable distinction among the different part-of-speech types, thus making it possible to distinguish among different past tenses or adverb types. This information is useful to detect when the authors are reporting causes or are talking about hypothetical future events.

  We captured the PoS using a mixed approach. We used the Stanford POS Tagger [34] to identify verbs, determinants, qualifying adjectives and ordinal adjectives. We then combined this Tagger with the usage of dictionaries in order to identify superlative adjectives and derogatory adjectives. In the case of adverbs, we distinguished among adverbs of time, mode, place, negation, affirmation and quantity, and adverbs that express doubt or desire. Determinants were classified as: (1) demonstrative, (2) possessive, (3) numeral, (4) exclamatory or (5) interrogative.

- **Punctuation and symbols (SYM)**. This set of dimensions capture typographical symbols used in communication. For example, sentence dividers, such as dots or semicolons, in order to capture the rhythm of the text, along with other general-purpose symbols, such as the *at* sign.

- **Twitter features (TWI)**. This set of features detect terms that are specific to Twitter, such as hashtags, mentions or hyperlinks. A hashtag

is used in Twitter to let allow the users to tag tweets, thus enabling other users to find related messages more easily. Hashtags have been used as a useful linguistic feature with which to identify cyberbullying [35]. The number of mentions, and their place within the text, can have a role in the communication. Mentions at the beginning of a tweet are considered responses to other tweets and can, therefore, be used to track conversations or to detect when users mention other users to draw their attention. We additionally applied heuristics in order to identify those cases in which the first mention of a tweet shows that it was directed towards a female, by comparing a list of common female names in Spanish to check whether the name appeared as a sub-pattern. Finally, we included a collection of words regarding Twitter argot. For example, users can use the expression "block" to prevent other users to looking their profiles[14].

- **Sociolinguistics (SLI)**. These features represent mentions in the text related to family, personal issues, affiliations, friendship, cognitive processes, among other features.

- **Topics (TOP)**. The aim of this category is to capture the topic of the statement. Some words regarding offensive language also appear in this category. As insults may reference sexuality, family, or physical appearance, it is possible to trace a relationship between the offensive words and these topics. For example, some insults in Spanish are feminine names of animals (such as *zorra*, or *perra*). Finally, we have captured the names of countries, capitals, and languages with the aim of detecting racism or other expressions of social exclusion.

- **Sentiment Lexicons (SEN)**. This category contains a lexicon regarding positive and negative emotions, including pleasure, anxiety, anger, sadness, and offensive words, which was compiled from various resources, such as the Spanish Opinion Lexicon (SOL) [36].

- **Stylometry (STY)**. This set of features contains metrics that capture the length of the tweets, along with the percentage of words and syllables in order to measure the readability of a text.

4.2. Evaluation

For the evaluation of the features, all the experiments were executed on the WEKA platform version 3.8.4 [37]. We selected three different machine learning classification algorithms: (1) Random-Forest (RF) as a decision tree classifier, and two variants of Support Vector Machines: (2) Sequential Minimal Optimization (SMO), and (3) Linear Support Vector Machines (LSVM). RF and SMO were selected because they have been applied to perform NLP tasks in the past [38, 39, 40] and because both were included among the 25 best

[14]The use of Anglicisms in social networks is common in the Spanish language

classifiers identified in [41]. LibLinear is also included in this study because it is faster for training larger datasets than SMO owing the lack of kernels, without a significant loss of accuracy [42].

- **Random-Forest (RF)**. Random-Forest is a decision tree algorithm that combines the predictions from multiple machine-learning models in order to improve the accuracy of the model executed separately [43]. The RF algorithm specifically creates sub-trees with the objective of generating less correlation among the features such that the accuracy of the models' learning is not over-fitted [44]. With regard to the hyper-parameterization, we configured RF without restrictions to the max-depth of the trees, a batch-size of 100 and 100 iterations. The other parameters used were the predefined parameters provided by WEKA.

- **Sequential Minimal Optimization (SMO)**. SMO is based on Support Vector Machines (SVM). This family of classifiers finds the optimum hyper-plane that distinctly separates the classes in the corpus [45]. SMO can handle non-linearly separable data, as may be the case of high-dimensionality problems, which are relevant for statistical models such as the BoW or char-grams. In the case of the hyper-parameterization, we configured SMO to use a polynomial kernel, with C equal 1 and a batch-size of 100. The other parameters used were the predefined parameters provided by WEKA.

- **LSVM**. We have employed LibLinear, which is an open source library with which to perform Logistic Regression and Linear Support Vector Machines [42]. LibLinear is efficient in large sparse data sets and allows different configurations. With regard to the hyper-parameterization, we applied the L1-regularized and L2-loss SVM type and a batch-size of 100. The other parameters used were the predefined parameters provided by WEKA.

Each model was trained following a 10-fold cross validation approach. The performance of each machine learning classifier was evaluated by comparing its accuracy (see Equation 1) and its standard deviation, since the whole corpus is balanced, as is each subset, signifying that no class can lead to the obtaining of high accuracy rates.

$$Accuracy = TP + TN/(TP + TN + FP + FN) \tag{1}$$

We evaluated our proposal by performing a baseline, applying a text representation model based on Bag of Words (BoW). BoW was selected owing to its ease of implementation and because it produces good results in NLP tasks [46]. However, BoW tends to create a high number of features, which consume many resources, as regards both time and memory. The features of each document with BoW were performed using the normalised version of the text. We additionally stripped the *URL_TOKEN* because this token can lead

to an over-fitted model. In order to calculate the percentage of unigrams in the documents, the Term Frequency-Inverse Document Frequency (TF-IDF) was obtained to measure the relevance of each feature within the corpus (see Equation 2 with the use of the normalised term frequency (see Equation 3) to prevent bias with common unigrams. We also perform performed an attribute selection to filter the most discriminatory unigrams and reduce the time required to train the model.

$$TFIDF = TF * IDF \tag{2}$$

$$TF = number_of_occurrences/number_of_grams \tag{3}$$

$$IDF = \log_2 corpus_size/documents_with_terms \tag{4}$$

Table 2: Accuracy and standard deviation of the baseline model (BoW) for VARW, SELA, DDSS and MisoCorpus-2020 by applying ten cross-fold validation

	RF		SMO		LSVM	
Feature set	ACC	SD	ACC	SD	ACC	SD
VARW	**78.930**	2.214	**78.524**	2.241	**80.053**	1.613
SELA	76.967	2.728	76.918	2.141	78.476	1.625
DDSS	74.734	1.795	74.003	2.568	77.698	1.742
MisoCorpus-2020	76.215	0.972	73.798	1.038	**77.060**	1.443

The results obtained for the baseline model for VARW, SELA, DDSS and MisoCorpus-2020 are shown in Table 2. As will be noted, the best result obtained for all the subsets (VARW, SELA, and DDSS) was attained with LSVM for the VARW subset, with an accuracy of 80.053% and a Standard Deviation of 1.613, whereas the worst accuracy was attained by SMO, with an accuracy of 74.003% and a Standard Deviation of 2.568 for the DDSS subset. With regard to the standard deviation, RF and SMO attain a high deviation with SELA and RF (2.728), which may indicate that the folds created to validate the baseline models discarded relevant unigrams among the folds. Despite the standard deviation of SELA, its accuracy is better than we expected. When we obtained first experimental results midway through the annotation process, the SELA results were worse than those of the other classifiers. Our first hypothesis was that cultural background and different variants of Spanish had a negative effect on the BoW model in comparison with other corpus, which shares the linguistic variant and cultural background.

Upon studying the results of the MisoCorpus-2020, it will be observed that LSVM achieves the best result, with an accuracy of 77.060% and a standard deviation of 1.443, and that it outperforms the accuracy of SMO (accuracy of 73.798% and Standard Deviation of 1.038) and RF (accuracy of 76.215% and

standard deviation of 0.972). With regard to the standard deviation of the baseline model with the MisoCorpus-2020, note that the deviation is always lower than the subsets of the corpus. This finding suggests that the unigrams obtained from each subset of the corpus conserve their semantic information. It will also be noted that the accuracy of SMO and LSVM is similar, or slightly worse, than the worst accuracy obtained by the same classifier for all the subsets. This finding does not, however, appear with RF, whose accuracy for the MisoCorpus-2020 is better than the worst result of the subsets.

About the performance of the classifiers, we observe that LSVM surpasses the rest of the classifiers for each subset and for the MisoCorpus-2020, in addition to having less standard deviation. The improvement in accuracy is especially relevant in the DDSS subset, for which LSVM increases the accuracy to 2.964% more than with RF and to 3.695% more than with SMO.

Once we had obtained the results of the baseline model, we compared them with those of the sentence embeddings model. We calculated the average of word embeddings (AWE) in each document in the corpus using the pre-trained word-embeddings in Spanish provided by fastText [47] with the intention of comparing the sparse and context-less text representation model with a dense and semantic text-information model. The results of the average number of word embeddings are shown in Table 3.

Table 3: Accuracy and standard deviation of AWE for VARW, SELA, DDSS and MisoCorpus-2020 evaluated with ten cross-fold validation

	RF		SMO		LSVM	
Feature set	ACC	SD	ACC	SD	ACC	SD
VARW	**82.092**	1.224	**84.886**	1.276	**84.480**	1.951
SELA	81.307	2.111	82.100	1.733	81.859	1.973
DDSS	79.063	2.009	81.360	1.586	81.148	1.745
MisoCorpus-2020	77.232	1.497	**81.020**	1.505	80.825	0.844

The data obtained by AWE (see Table 3) improves the results of the baseline model for each subset in the corpus. Specifically, the SMO classifier obtains the best accuracies, with 84.886%, 82.100%, and 81.360% for VARW, SELA and DDSS, respectively. Moreover, the standard deviation of SMO is lower than the standard deviation of the other classifiers, with the exception of VARW. Note that the standard deviation is higher in the RF model than in the other classifiers.

Upon studying of AWE when applied to the MisoCorpus-2020, it will be observed that SMO significantly improves the accuracy of the baseline model (81.020 vs. 73.798), but with a major standard deviation (1.505 vs. 1.038). Of the three classifiers, LSVM attains a significantly lower standard deviation (0.844) than SMO (1.505) and RF (1.497). However, despite the increase in the standard deviation, these results suggest that it is possible to improve the identification of misogyny with a reduced set of features. However, word

embedding approaches hinder the ability to understand why this works, since documents are coded as vectors in which it is difficult to perform reverse engineering in order to understand the classification process. We, therefore, performed an evaluation of the corpus with LF to discover whether there were any specific linguistic phenomena related to misogynous texts and whether the results would improve.

The results of the evaluation of LF are shown in Table 4. The first aspect that will be noted is that LF loses accuracy when compared to the accuracy of AWE. The LF model obtains an accuracy of 82.403% for the VARW corpus, whereas SMO attains an accuracy of 84.886% with AWE (see Table 3). In the case of the SELA subset, LF attains an accuracy of 81.740% with the RF classifier, whereas AWE attains an accuracy of 82.100% with SMO (see Table 3). Finally, in the DDSS corpus, the LF model obtains an accuracy of 79.245% with LSVM, whereas AWE attains an accuracy of 81.360% with SMO (see Table 3). In the case of the MisoCorpus-2020, the best result attained by the LF model underwent a slight decrease to an accuracy of 79.263% with the LSVM classifier, whereas the AWE model attained an accuracy of 81.020% with SMO (see Table 3).

LF attains the worst standard deviation for all the classifiers with the MisoCorpus-2020, doubling the standard deviation attained with LSVM from 1.658 to 0.844 (see Table 3. This behaviour can also be appreciated with the subsets, as SMO attains a standard deviation that exceeds 2% on all the subsets. The worse standard deviation for LF suggests that the cross-fold validation omits some relevant linguistic features.

When compared with the baseline model (see Table 2), the accuracy is higher with LF for all the subsets, whereas the standard deviation is also lower, with the exception of DDSS with RF. LF also surpasses the baseline model for the MisoCorpus-2020, attaining the maximum difference, with an accuracy of 5.14% with the SMO model. However, the standard deviation is worse with LF than with the baseline model.

Table 4: Accuracy and standard deviation of LF features for VARW, SELA, DDSS and MisoCorpus-2020 evaluated with ten cross-fold validation

	RF		SMO		LSVM	
Feature set	ACC	SD	ACC	SD	ACC	SD
VARW	81.112	1.040	**82.403**	2.042	**82.283**	1.535
SELA	**81.740**	1.949	80.057	2.067	81.115	1.865
DDSS	77.613	2.190	77.976	2.056	79.245	1.078
MisoCorpus-2020	79.237	1.757	78.938	1.631	**79.263**	1.658

In view of these results, we consider that LF sacrifices accuracy in favour of interpretability with respect to AWE. However, the results obtained by LF are quite a lot better than those obtained by the baseline model. We then combined AWE and LF with the objective of improving the overall accuracy of the system.

The results of this model are shown in Table 5.

Table 5: Accuracy and standard deviation of AWE and LF features for VARW, SELA, DDSS and MisoCorpus-2020 when evaluated with ten cross-fold validation

	RF		SMO		LSVM	
Name	ACC	SD	ACC	SD	ACC	SD
VARW	**82.092**	1.224	84.886	1.276	**84.480**	1.951
SELA	81.307	2.111	**85.175**	1.450	83.734	1.915
DDSS	78.912	2.690	81.208	1.766	80.755	1.113
MisoCorpus-2020	79.302	1.497	**85.175**	1.450	82.882	1.291

The results shown in Table 5 indicate that SMO attains the best results for all the subsets, with an accuracy of 85.175% and a standard deviation of 1.450 for the SELA dataset, whereas the worst result is attained by RF, with an accuracy of 78.912% and a standard deviation of 2.690 for the DDSS dataset. In the case of the MisoCorpus-2020, the best result is attained with SMO, with an accuracy of 85.175% and a standard deviation of 1.450.

In order to ease the comparison of the results across each model and each feature set, Table 6 shows all the accuracy values for the VARW, SELA and DDSS subsets, and for MisoCorpus-2020 (SMC-2020).

Table 6: Resume

Classifier	Model	VARW	SELA	DDSS	SMC-2020
RF	BoW	78.930	76.967	74.734	76.215
	AWE	**82.092**	**81.307**	**79.063**	77.232
	LF	81.112	81.740	77.613	79.237
	AWE+LF	82.092	81.307	78.912	**79.302**
SMO	BoW	78.524	76.918	74.003	73.798
	AWE	**84.886**	82.100	**81.360**	81.020
	LF	82.403	80.057	77.976	78.938
	AWE+LF	84.886	**85.175**	81.208	**85.175**
LSVM	BoW	80.053	78.476	77.698	77.060
	AWE	**84.480**	**81.859**	**81.148**	80.825
	LF	82.283	81.115	79.245	79.263
	AWE+LF	84.480	83.734	80.755	**82.882**

As can be observed in Table 6, AWE+LF obtains better results than the other approaches in every classifier algorithm for the MisoCorpus-2020 corpus, with the best results being obtained when employing the SMO classifier. This increase in accuracy suggests that the features from LF and AWE are complementary and do not provide redundant or contradictory information. However, AWE obtains slightly better results for the DDSS dataset. As this

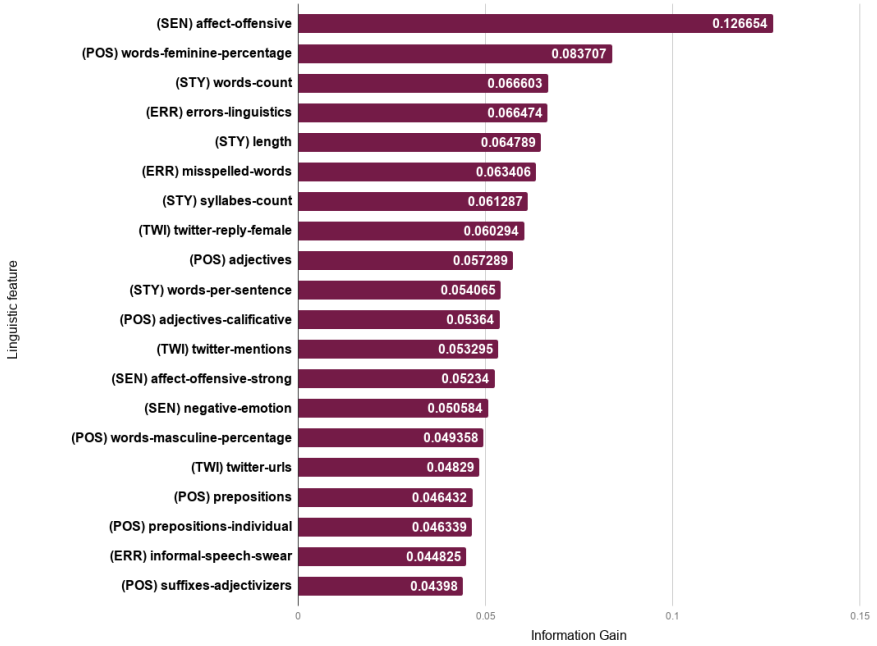
dataset contains fewer tweets and has different topics, the LF approach cannot provide useful information when the topic is more heterogeneous.

5. Discussion

The experiments performed confirmed that the combination of Linguistic Features (LF) and Average of Word-Embeddings (AWE) improved the detection of misogyny and oppressive speech. The performance achieved outperformed the baseline model based on BoW, as also occurred with AWE and LF, witch attained accuracy of 85.175% and a standard deviation of 1.450. This result suggests that these sets of features are complementary and some of the linguistic phenomena related to oppressive speech cannot be represented merely as statistics regarding words.

We obtained the most discerning linguistic features by calculating the weight by which each linguistic feature contributes to the class prediction. This information is expressed as Information Gain, and this metric is usually employed by ensemble methods such as RF to decide when new branches need to be created. The top 20 features with major information gain appears in Figure 4.

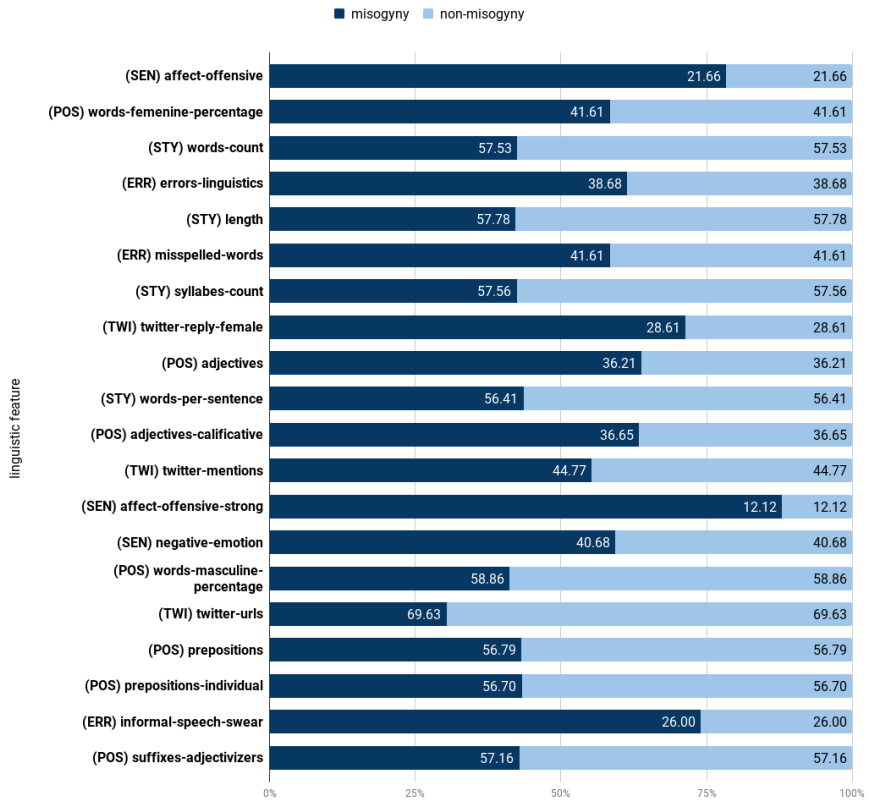
Figure 4: Information gain of the 20 best linguistic features



Note that the feature that provides the most information gain is offensive language *((SEN) affect-offensive)*, with 0.126654. This feature includes offensive

words and expressions in Spanish and which appear to the greatest extent in misogynist statements. The second linguistic feature to provide major information gain is *POS-words-feminine-percentage* with 0.083707. This category refers to the number of words that are grammatically feminine. Some linguistic features that provide similar information gain are related to (1) stylometry, such as the number of words or the length of the tweet, and (2) grammatical errors, such as mistakes in writing *((ERR) linguistic-errors)* or the percentage of misspelled words (*misspelled-words*). Features related to Twitter are also important, of which we can highlight the number of mentions. Mentions are used in Twitter to reply to other users' messages (when they appear at the beginning of the conversation), or can be added to include other users in the conversation. We counted the number of mentions *(TWI) twitter-mentions*, and attempted to deduce whether the first mention in the tweets referred a to a woman *(TWI) twitter-reply-female*.

Figure 5: Stacked mean of the best 20 linguistic features



We then calculated the stacked mean of the 20 best linguistic features for the MisoCorpus-2020 in order to analyse which linguistic features appear in documents labelled as *misogyny* or *not-misogyny*. As not all the linguistic features appear in the same degree in the corpus, the stacked mean normalises each value to a value between 0 and 100 so in this easy to analyse when a linguistic feature appears more often in misogyny utterances or not. In this respect, a value near 50% indicates that the linguistic feature appears with the same frequency in tweets labelled as *misogyny* and tweets labelled as *non-misogyny*. The results are shown in Figure 5.

After an analysis of these features, we attained the following insights:

- **Offensive language**. The most discerning linguistic feature is *affect-offensive*. Offensive language was represented by dividing the linguistic features into: (1) *affect-offensive-soft* which includes weak swear words or other slang used in Spanish for cursing in general, and (2) *affect-offensive-strong*, which includes words and phrases whose intention is to intimidate another interlocutor. We observed that *affect-offensive-strong* provides more than double the amount of information gain than the complementary discerning feature. Moreover, some misogynist specific topics, such as *violence menaces* and *disparagement* appear as discerning features but, mostly in tweets labelled as *not misogyny*.

- **Grammatical gender** The percentage of words that are grammatically feminine or masculine are discerning linguistic features. These categories can be used effectively for languages with grammatical gender for identifying easily messages towards a specific gender. In other cases, researches have made use of dictionaries of proper names to identify the gender of the participants of a conversation. It is, however, difficult to obtain this information from social networks, where people are not required to use their real names. Moreover, the grammatical gender in Spanish has caused debate about if Spanish can be considered as sexist language [48]. The grammatical gender in Spanish has led to debates as to whether Spanish can be considered as a sexist language [48]. Nevertheless, some words in Spanish have a figurative meaning according to their gender. For example, *zorro* (a male fox) can be used as qualifying adjective to detonate a cunning person, while its feminine equivalent *zorra* (a female fox) can be used to denote a female prostitute[15]. Another word of this nature is *Perra* (female dog).

- **Grammar and spelling mistakes** The percentage of *misspelled words*, *words-with-accents*, and *words-in-dictionary-avg*, suggests that mastering language correctly can be a relevant factor to take into consideration when identifying misogynous messages. Moreover, *informal-speech-collocations*, which are redundant expressions, suggest that users do not take enough

[15] https://dle.rae.es/zorro. Point 7 (Accessed: April 28, 2020)

time to consider and prepare their arguments, and their impulsive speaking, therefore, contributes to the detection of misogynous statements.

- **Part of Speech**. With regard to the percentage of words according to their grammatical type, the percentage of qualifying adjectives is the most discerning feature. In this case, these types of adjectives, which are used to describe things, have a strong correlation with *offensive-language*. The percentage of adverbs, which include expressions such as *además* (besides), or locutions such as *en lugar de* (instead of), additionally indicate that the user is dealing with two ideas in the same topic, using one statement to cast blame and other to suggest an alternative.

- **Punctuation and symbols**. In the case of symbols, those features that provide major information gain are the *char-colon*, *char-close-exclamation* and the *char-quotation-close*. The high percentage of colons in tweets labelled as *not misogyny* indicate that the author has a more direct writing style. The percentage of exclamation marks is used to add emphasis to a statement. This feature is sometimes correlated with the percentage of uppercase words, but, this correlation does not appear in our results. Finally, *char-quotation-close* is used to quote other people's comments and can be used to mock an argument by making use of fallacies based on the straw man [49]. The *char-pipe* symbol is principally used in tweets written by news sites. As these tweets were classified differently and labelled as not misogyny, they can be safely discarded.

- **Twitter features**. The number of mentions and URLs in the twitter posts can be used as a feature to determine misogynous comments. In this respect, the attacker may wish to increase visibility in order to draw the attention of other attackers. This behaviour also appears in other cases of harassment, such as cyberbullying [50].

In order to compare how the cultural background affects misogyny identification, we conduct an experiment that consisted of dividing the SELA corpus into two subsets: (1) a training set with tweets from Spain and (2) a test set with tweets from Latin America. We specifically extracted 2976 tweets from Spain (1488 annotated as misogyny and 1488 annotated as not misogyny) and 984 tweets from Latin America (492 annotated as misogyny and 492 annotated as not misogyny). Note that the sum of these tweets (2976 and 987) is not exactly the total of the size of the SELA corpus, which is composed of 4162 tweets. The reason for this difference is because we discarded those tweets whose origin we did not know how to properly identify. The results of the accuracy of each classifier are shown in Table 7.

The results in Table 7 show that AWE+LF outperforms the results of AWE and LF separately. In addition, AWE always attains the worst results, no matter which classifier is used. However, the usage of LF improves the results of AWE by 11.7886% for RF, 8.6382% for SMO, and 11.0772% for LSVM. AWE+LF improved these results significantly when SMO was applied.

Table 7: Comparison of the accuracy between European Spanish and Latin American Spanish when applying AWE, LF, and AWE+LF

Feature set	RF	SMO	LSVM
AWE	57.3171	61.0772	61.3821
LF	69.1057	69.7154	72.4593
AWE+LF	**69.4106**	**75.1016**	**73.1707**

However, the accuracy did not improve greatly with respect to RF and LSVM. As a conclusion, these results show that misogyny shares common features from different cultural backgrounds, including linguistic features and other features related to the usage of Twitter.

5.1. Comparison with other datasets

We performed a further analysis in which we compared our model (AWE+LF) with other available datasets. We first evaluated AWE+LF in task A of the AMI 2018 task [13]. This task consisted of a binary classification with which to identify misogynous statements. The results are shown in Table 8, in which it is possible to compare our AWE+LF–trained proposal with two machine learning classifiers: LSVM and SMO. These classifiers were selected because LVSM was similar to the classifier used by those participants in the task who achieved the best results [51, 52, 53] and SMO was the machine learning classifier that achieved the best accuracy for the MisoCorpus-2020. The results include the baseline of Task A of the AMI 2018 and the participants' five best runs.

Table 8: Comparison of our AWE+LF model (Average Word Embeddings with Linguistic Features) with the best five runs of Task A and the baseline of the AMI 2018 Task A with the Spanish dataset

Run	ACC
AWE+LF (LSVM)	**81.5217**
14-exlab.c.run3 [51]	81.4681
JoseSebastian.c.run1 [52]	81.4681
SB.c.run4 [53]	81.3478
14-exlab.c.run1 [51]	81.2274
14-exlab.c.run2 [51]	81.2274
AWE+LF (SMO)	79.2271
AMI-BASELINE [13]	76.7750

As the results in Table 8 show, our proposal slightly outperforms the best result achieved by both Pamungkas et al. [51], and Canós [52] with an LSVM. However, our accuracy is worse with SMO than the five best results, but still exceeds the baseline accuracy. With regard to the features used, Pamungkas et al. [51] represent the documents with a combination of n-grams (unigrams,

bigrams, and trigrams) with linguistic features that count swearwords and sexist slurs. They additionally identify topics related to genitalia, prostitution or homosexuality. In the case of Twitter features, they identify hashtags, emojis, and hyperlinks. Similarly, Canós [52] used a BoW model based on TF-IDF composed of all the tweets in the AMI corpus [13]. During the preprocessing stage, he identified expressive markers, such as multiple concatenated exclamation and question marks, and combined some hashtags under specific tokens. However, he replaced all the mentions in the texts, but did not handle the specific case of the first mention as we do. The next best result was achieved by [53], who employed linguistic features related to sexuality, profanity and femininity, among others, along with twitter features such as *hashtags* and the presence of sentiment lexicons.

We then compared our model with the corpus provided for task B of the AMI dataset. This task consisted of identifying two misogynous features: (1) the identification of misogynous traits (stereotype, dominance, derailing, sexual harassment, discredit), and (2) the identification of the target, referred to as active or passive. In this case, the comparison has been made by using the Macro F1-score formula. The results are shown in Table 9.

Table 9: Comparison of our model with the results attained by participants in AMI (Task B, with the Spanish dataset) using the Macro Avg for misogynous traits (derailing (der), discredit (dis), dominance (dom), sexual harassment (sex), and stereotype (ste))

Run	der	dis	dom	sex	ste
14-exlab.c.run2 [51]	0.0000	70.2886	37.8788	**53.0120**	8.3333
14-exlab.c.run3 [51]	0.0000	**70.6081**	37.5940	51.7647	8.3333
14-exlab.c.run4 [51]	0.0000	70.5882	37.5940	51.1628	8.3333
SB.c.run4 [53]	0.0000	68.0702	**39.6396**	43.1818	14.2857
14-exlab.c.run1 [51]	0.0000	70.1695	37.8788	53.0120	8.6957
JoseSebastian.c.run1 [52]	0.0000	64.8464	33.3333	52.8736	10.5263
AMI-BASELINE [13]	0.0000	64.9660	28.5714	38.9610	8.3333
AWE+LF (SMO)	0.0000	**62.8466**	**29.5652**	44.1860	0.0000
AWE+LF (LSVM)	0.0000	57.2534	23.1579	**48.8373**	**16.0000**

Despite the good results attained in Task A in AMI, our proposal improved only the baseline results of Task B for tweets annotated as *dominance* and *sexual harassment*. However, our approach attained lower score for *derailing*. In the case of *stereotype*, our approach obtained the best score when using an LSVM. These results indicate that our proposal is good as regards identifying misogyny as a binary classification, but that it fails in the case of multi-class classification problems. When studying the statistics of the corpus, it will be noted that derailing (the ability to redirect the conversation towards something that is more comfortable for men) and stereotypes (an oversimplified image or idea of women) do not contain so many examples (0.6% for derailing and 4.464% for stereotype for the whole training set). Nor do the best runs correctly identify tweets. When viewing the results for discredit (the slurring of women with no

other intention), which is the trait that appears most frequently in the corpus (29.6689% for the whole training set), we attained an F-Measure of 62.8466% with SMO, which is lower than the AMI Baseline (64.9660% of F1-Measure).

We then evaluated our approach with the second part of Task B, which consisted of identifying the target. We again evaluated our approach using the SMO and LSVM classifiers. These results are shown in Table 10. In this case, we achieved the best F-Measure with SMO (76.6304% for active, and 20.0000% for passive). As our passive score was lower than the baseline, this shows that our proposal mistakes tweets directed towards the general public for non-misogynistic tweets.

Table 10: Comparison of our model with the results attained by participants in AMI (Task B, with the Spanish dataset) using the Macro Avg for misogynous target (active, passive)

Run	active	passive
14-exlab.c.run2 [51]	81.6062	29.0323
14-exlab.c.run3 [51]	**82.0051**	29.0323
14-exlab.c.run4 [51]	81.5857	29.0323
SB.c.run4 [53]	79.8387	**30.5085**
14-exlab.c.run1 [51]	81.6537	26.6667
JoseSebastian.c.run1 [52]	81.3245	27.1186
AMI-BASELINE [13]	77.0861	30.3030
AWE+LF (LSVM)	76.0962	14.5454
AWE+LF (SMO)	**76.6304**	**20.0000**

Finally, we calculated the Macro Average F-Measure from the results attained by the Macro F-Measure obtained by identifying the traits (see Table 9) and identifying the target (see Table 10). As will be observed (see Table 11), the Macro Avg. attained by our proposal does not improve the baseline of Task B with either SMO or LSVM. However, in the specific case of SMO, the difference between the baseline results attained by our proposal and those attained by the other is small (0.075).

We then compared our proposal with Task A of HatEval 2019 [20]. In this case, the corpus was composed of tweets with hate speech messages directed towards immigrants and women in a binary classification, in which tweets were rated as hateful or non-hateful. The results are shown in Table 12.

As the results in Table 12 show, our proposal increases both the maximum result and the baseline attained in Task A, regardless of the classifier used, with the best Macro-Averaged F1-score of 75.4505% being attained with SMO. This result suggests that the identification of misogyny and hate speech are highly related. It is important to stress that our models were trained only with the HatEval training dataset that included tweets in which the targets were women or immigrants.

As a conclusion regarding the comparison with previous corpora and tasks, our approach AWE+LF works well with binary classification, obtaining good results, but it does not obtain good results as regards multi-class classification

Table 11: Comparison of our model with the results attained by participants in AMI (Task B, with the Spanish dataset) using the Macro Avg., and the Macro F-Measure for category, and the Macro F-Measure for target

Run	Macro Avg.	Traits	Target
14-exlab.c.run2 [51]	**44.6109**	33.9026	55.3192
14-exlab.c.run3 [51]	44.5894	33.6600	**55.5187**
14-exlab.c.run4 [51]	44.4223	33.5357	55.3090
SB.c.run4 [53]	44.1045	33.0355	55.1736
14-exlab.c.run1 [51]	44.0557	**33.9512**	54.1602
JoseSebastian.c.run1 [52]	43.2807	32.3398	54.2216
AMI-BASELINE [13]	40.9185	28.1424	53.6946
AWE+LF (SMO)	**40.8435**	27.3196	**54.3673**
AWE+LF (LSVM)	38.6825	**29.0497**	48.3152

Table 12: Comparison of our model with the results attained by participants in HatEval (Task A) with the Spanish dataset)

	Macro-Averaged F1 score
AWE+LF (SMO)	**75.4505**
AWE+LF (LSVM)	73.3041
Max	73.0000
SVG HatEval Baseline	70.1000
Min	49.3000

problems.

6. Conclusions and further work

In this paper, we present and evaluate a model for misogyny identification for the Spanish language based on Average Word Embeddings and Linguistic Features (AWE+LF) using a new corpus compiled with tweets in Spanish. The model obtains an overall accuracy of *85.175%* with SMO which outperforms a baseline model based on BoW, and the models based on linguistic features (LF) and the Average of Word Embeddings (AWE) separately. We additionally used our model to the evaluate three subsets of the corpus: oppressive messages towards relevant women (VARW); linguistic differences between the Spanish spoken in Europe and that spoken in Latin America (SELA), and with texts concerning general traits related to misogyny such as discredit, dominance, sexual harassment or stereotyping (DDSS). We have also compared our approach with available datasets regarding misogyny and hate speech towards women in Spanish. This analysis shows good results as regards binary classification tasks, although the results obtained in multiclass classification problems could be improved.

We expect our contribution will improve current technologies in Social Computing providing automatic misogyny detectors that could improve safety in social networks. More concretely, it could be applied in smart cities applications that can benefit from different social aspects such as equality, inclusion or respect [54].

As further work, we shall continue to enlarge the MisoCorpus-2020. First, regarding the differences between dialects, we have scheduled to organisation of labelling processes with people from different Latin America countries, signifying the labelling process will performed by native speakers. We shall additionally continue with the tweet-labelling process, but this will be done by more annotators and with a heterogeneous group of the population, including people of different sexes and age ranges. This will allow us to attain a better understanding of how misogyny is perceived and thus detect the features that make the automatic identification of misogyny difficult. We are designing a tool that is capable of monitoring Twitter as regards real-time tweets concerning new events, and intend to evaluate the quality of our proposal with a new data set. Finally, with regard to the low score attained in Task B of AMI, we plan to tag the MisoCorpus-2020 with the same labels described in AMI [13] in order to gain a greater understanding of misogynistic traits. Furthermore, new models for the improvement of multi-class classification problems will be developed in order to overcome the limitations of the AWE+LF approach.

Finally, current contextualized word embeddings approaches such as BERT [55] and ELMo [56] have been widely applied to multiple NLP tasks such as emotion detection [57] or suggestion mining [58]. It is also planned to evaluate these technologies for misogyny identification in order to improve current results of this research.

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